SESSION 7 – LOGISTIC REGRESSION WITH POLYNOMIAL INPUTS

(Non-Linear Classification)

🎯 **Topic**

- Why linear boundary fails
- Non-linear classification
- Polynomial features in classification
- Decision boundary (curve)
- Practical implementation

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1. 🎯 Learning Objectives

By the end of this session, students will be able to:

- Understand why **linear classification is not always sufficient**
- Learn how to handle **non-linear data**
- Apply **Polynomial Features in Logistic Regression**
- Visualize **curved decision boundary**
- Build **advanced classification model**

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2. 🔁 Quick Recap (Connect Previous Session)

Start class like this:

"In the last session, we used multiple inputs in Logistic Regression.
But all those models created a straight decision boundary.
Today we will learn what to do when the data is not linearly separable."

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3. 📖 Problem: Linear Boundary Limitation

Example

Imagine dataset like this:

- Class 0 → inner circle
- Class 1 → outer circle

👉 Can a straight line separate them?

❌ No

Conclusion

"Linear Logistic Regression fails when data is non-linear."

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4. 📊 Solution: Polynomial Features

Idea

Transform input features into higher powers:

- $x \rightarrow x^2$
- $x \rightarrow x^3$
- combine features

Teaching Line

"We don't change the algorithm, we change the input."

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5. 📊 How It Works

Step 1: Original Input

$$x$$

Step 2: Transform Input

$$x, x^2, x^3$$

Step 3: Apply Logistic Regression

👉 Now model can create **curved boundary**

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6. 📈 Decision Boundary (Important)

Linear Case:

- Straight line

Polynomial Case:

- Curve / circle / complex shape

Teaching Line

"Polynomial features allow Logistic Regression to draw curves instead of straight lines."

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7. 📊 Real-Life Example

Example: Loan Approval

Decision depends on:

- Income
- Age

👉 Boundary is not always linear:

- Young + high income → Approved
- Old + low income → Rejected

👉 Complex patterns → need polynomial

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8. 📊 Key Advantages

✅ Handles non-linear data
✅ Improves model accuracy
✅ Still uses Logistic Regression
✅ Flexible decision boundaries

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9. ⚠️ Important Warning (Overfitting)

If degree is too high:

- Model becomes too complex
- Fits noise instead of pattern

Teaching Line

"Higher degree is not always better — it can cause overfitting."

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10. 🎯 Interview Concepts

Q1: Why use polynomial in classification?

👉 To handle non-linear data.

Q2: Does Logistic Regression change?

👉 No, only input features change.

Q3: What is decision boundary?

👉 Boundary separating classes.

Q4: What is overfitting?

👉 Model learns noise instead of pattern.

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11. 💻 PRACTICAL (YOUR FORMAT)

🎯 Objective

Classify data where:

- Linear boundary fails
- Polynomial boundary works

Output Interpretation

Explain clearly:

Observation:

- Linear model struggles
- Polynomial model performs better

👉 Conclusion:

"Polynomial transformation helps in capturing complex patterns"